Residual $SO(4,2)$ Cohomology of Free Weyl Gravity on the Conformal Cylinder: Cartan Homotopy, Weyl-Square Classes, and Their Residual Pairing

Research-programme draft
Authorship and attribution to be finalized after external review

Draft for external review, July 2026

Abstract

Treating all fifteen cylinder conformal transformations, including compact time translation D, as residual gauge transformations, we compute a minimal absolute $SO(4,2)$ Chevalley–Eilenberg complex for free four-dimensional Weyl gravity on $\mathbb{R} \times S^3$. Its coefficient module is the parity-complete on-shell Weyl-curvature Fock module. The standard invariant one-particle form is indefinite, with cylinder signs $+I_E \oplus (-I_A) \oplus (-I_L)$.

The key mechanism is the Cartan homotopy formula. The residual differential obeys $d\iota_D + \iota_D d = \mathcal{L}_D$, so every subcomplex of nonzero total compact degree is contractible. In the Hamada-centered four-ghost polarization, the selected absolute degree has ghost compact degree bounded below by -4, and only matter weights at most four can contribute. The vacuum and one-particle sectors are acyclic, while a cutoff-complete absolute calculation in the lowest two-particle sector gives

$$H^4_{\text{res,min}} = \text{span}\{[W_+^2], [W_-^2]\}, \qquad G_{\text{res}} = I_2.$$

Here $G_{\text{res}} = I_2$ is the induced Hermitian complementary-degree CE pairing in the displayed residual basis. The survivors are ghost-dressed vertex or deformation classes, not one-particle graviton states. Descent identifies their parity-even and parity-odd combinations with $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$ and $C_{\mu\nu\rho\sigma}\widetilde{C}^{\mu\nu\rho\sigma}$. The first varies to the Bach equation; the second is the Pontryagin boundary direction. After quotienting variationally trivial bulk functionals, the local dynamical deformation space is one-dimensional and has residual Gram matrix I_1.

This algebraic residual theorem is exact once the on-shell Weyl module, the Hamada ghost polarization, and full residual conformal gauging are assumed. The last assumption is a physical choice: if D is retained as a Hamiltonian or boundary charge, the contraction is not a physical quotient. The flat adjoint BGG resolution and the topology of $\mathbb{R} \times S^3$ additionally give an all-level smooth isomorphism between the Bach quotient and a geometrically defined on-shell curvature system. Three further steps remain conditional: exactness on the algebraic positive-energy cylinder module, the full equivariant cyclic BV transfer with its antifield rows, and derivation of the centered BFV polarization from the chosen closed-universe reduction. Any analytic completion requires additional control. We claim neither a positive graviton Fock space nor quantum anomaly freedom.

Contents

1 Introduction

Pure conformal gravity in four dimensions is defined, up to an overall coupling and the Euler density, by the Weyl-square action

$$S_W[g] = -\alpha_g \int_M d^4x \sqrt{-g}\, C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}. \tag{1}$$

Its fourth-order linearized equation has more solutions than the Einstein equation, and a conventional local reduction produces an indefinite oscillator form. This familiar observation does not by itself determine the state space after all gauge constraints have been imposed. On a compact spatial slice, background conformal symmetries are reducibilities of the local Diff×Weyl gauge system. Treating those residual transformations as gauge introduces a second, finite-dimensional reduction which is absent from a naive oscillator count. It is also a substantive physical choice. In a Dirac or background-independence interpretation of a closed universe, all residual conformal transformations can be quotiented. In a fixed cylinder conformal field theory, by contrast, D is ordinarily the Hamiltonian and the conformal group acts globally. This paper computes the first choice; it does not assert that strict pure-Weyl gravity uniquely selects it.

The organizing question is therefore more basic than a sign assignment: before asking whether a gravitational mode has positive or negative norm, one must first determine whether it survives the full gauge reduction being imposed. The exact answer below concerns the residual reduction. The smooth metric-to-curvature bridge is also proved globally; the passage to the algebraic mode complex and then to the complete local-plus-residual BV/BFV theory remains conditional.

The present paper isolates that free classical problem on

$$M = \mathbb{R} \times S^3, \qquad \bar{g} = -dt^2 + d\Omega_3^2, \tag{2}$$

with unit cylinder radius. Three spaces must be distinguished:

$$\begin{gathered} \text{local oscillator cohomology} \supseteq \text{classically integrable perturbations}, \\ \text{classically integrable perturbations} \longrightarrow \text{residual-BRST classes}. \end{gathered} \tag{3}$$

The first contains the familiar Einstein, vector-descendant, and logarithmic or higher-derivative towers. The last is obtained only after the residual $SO(4,2)$ constraints and their ghosts are included. A linearized oscillator need not survive this final reduction.

Our result is most naturally stated in the language of residual vertex cohomology. The Hamada ghost polarization supplies a top-degree four-ghost factor of compact degree -4. It pairs with scalar matter primaries of weight four. The complete centered residual cohomology consists of two chiral Weyl-square classes, and their normalized residual CE Gram matrix is positive. Their parity combinations are the dynamical Weyl-square density and the topological Pontryagin density. Thus the positive matrix is attached to classical vertex or theory-space directions. It is not a positive norm on a propagating graviton Fock space; indeed, the one-particle residual cohomology vanishes.

Relation to previous work. The conformal deformation detour complex on obstruction-flat backgrounds was constructed by Branson and Gover [1]; its formal self-adjointness and related Yang–Mills detour constructions were developed in [3]. Gover and Peterson give the oriented four-dimensional deformation sequence, while Čap proves that the flat BGG sequence is a fine resolution and computes the same cohomology as the flat twisted de Rham complex [2, 5].

Cylinder harmonics, oscillator charges, and primary building blocks were developed by Hamada and Horata and by Hamada [8, 9]. In the broader conformal-mode/Riegert programme, Hamada identified the residual diffeomorphisms with cylinder conformal transformations [10], constructed the fifteen-generator residual BRST charge, used the product of four proper-conformal ghosts as the centered ghost vacuum, derived the scalar weight-four condition, and displayed a Weyl-square BRST state [11]. None of those ingredients is claimed as new here. We use the harmonic and residual-algebra conventions, but we do not import the Riegert field or the quantum claims of that model. The Weyl-curvature module and its character are consistent with the general conformal higher-spin description and partition-function literature [12, 13]. Finally, the identification of C^2 as the unique local four-derivative self-interaction of one linear conformal graviton, under the assumptions stated there, is due to Boulanger and Henneaux [14].

The defensible new result is narrower and more concrete. Starting with the parity-complete pure-Weyl coefficient module, we compute the complete minimal *absolute* residual complex in the centered degree. We prove that the vacuum and one-particle sectors vanish, that precisely two two-particle classes survive after the incoming-image test, and that their normalized Hermitian residual CE Gram matrix is I_2. The standard Cartan homotopy makes the particle-number inventory exhaustive, and the Euler–Lagrange map splits the two positive directions into one dynamical and one topological class.

Claim boundary. Four statements must remain separate.

1. The *algebraic residual theorem* is exact for the minimal absolute $SO(4,2)$ Chevalley–Eilenberg complex built on the on-shell Weyl module, the stated centered ghost polarization, and full residual conformal gauging.

2. The smooth complexified metric-to-geometric-curvature bridge is an exact global theorem. Restricting it to the D-finite cylinder category and identifying its target with the previously constructed $E/A/L$ coefficient module require an explicit algebraic intertwiner or equivariant BGG homotopy. Smooth equivariance of quotient classes alone does not supply finite-mode metric representatives.

3. The complete strict-pure-Weyl local-plus-residual BV statement remains conditional on a full equivariant cyclic zero-mode transfer with the relevant ghost and antifield rows, and on a residual BFV choice which induces the centered ghost insertion and its normalization. These remaining hypotheses are displayed rather than hidden.

4. Calling the resulting classes physical states additionally requires a boundary and gauge principle which makes all fifteen conformal generators, including D, gauge rather than global charges. We call them *candidate physical classes under full residual conformal gauging.*

No statement in this paper establishes quantum BRST nilpotency, anomaly cancellation, interacting pseudo-unitarity, or a positive asymptotic particle space.

The local-to-residual bridge belongs to the general subject of local BRST cohomology with antifields, gauge fixing, and descent [15, 17]. Metsaev's ordinary-derivative BRST–BV construction gives a concrete realization of conformal fields and their conformal algebra on fields and antifields [18]. Dneprov and Grigoriev construct a minimal nonlinear Weyl-jet model and compatible presymplectic structure [19]. Their curvature-dependent ghost transformations make that model a valuable comparison, but not an immediate proof that the *free* transferred residual differential is strict. Neither work supplies the global cylinder zero-mode normalization used here.

2 The free local pure-Weyl complex

2.1 Gauge, curvature, and equation maps

Let $h_{\mu\nu}$ be a metric perturbation and let (ξ^μ, σ) be an infinitesimal Diff$\times$Weyl parameter. The full linear gauge map is

$$K(\xi,\sigma)_{\mu\nu} = \mathcal{L}_\xi \bar{g}_{\mu\nu} + 2\sigma\bar{g}_{\mu\nu} = 2\bar{\nabla}_{(\mu}\xi_{\nu)} + 2\sigma\bar{g}_{\mu\nu}. \tag{4}$$

After quotienting the trace, it becomes the conformal Killing operator

$$(K_0\xi)_{\mu\nu} = 2\bar{\nabla}_{(\mu}\xi_{\nu)} - \frac{1}{2}\bar{g}_{\mu\nu}\bar{\nabla}\cdot\xi. \tag{5}$$

Let

$$C_1 h = \frac{d}{d\varepsilon}C[\bar{g} + \varepsilon h]\bigg|_{\varepsilon=0} \tag{6}$$

be the linearized Weyl-curvature map, and let B_{lin} denote the action-normalized linearized Bach operator. Formal adjoints use the spacetime action pairing on a domain for which temporal and spatial boundary terms vanish.

The conformal cylinder is conformally flat. Naturality and conformal covariance therefore give the exact local identity

$$\boxed{C_1 K = 0} \tag{7}$$

for arbitrary local gauge parameters, not only for the fifteen residual conformal Killing parameters.

Proposition 2.1 (Action-normalized factorization)**.** *Use the repository normalization*

$$S_{\text{red}}[g] = \int \sqrt{-g}\left(R_{\mu\nu}R^{\mu\nu} - \frac{1}{3}R^2\right) = \frac{1}{2}\int \sqrt{-g}\,(C^2 - E_4). \tag{8}$$

On a conformally flat background, modulo the Euler boundary term,

$$\boxed{B_{\text{lin}} = C_1^\sharp C_1.} \tag{9}$$

Consequently,

$$B_{\text{lin}}K = 0, \qquad K^\sharp B_{\text{lin}} = 0, \qquad B_{\text{lin}}^\sharp = B_{\text{lin}}. \tag{10}$$

Proof. Since $C[\bar{g}] = 0$, the mixed second variation of (8) is

$$\delta_h\delta_k S_{\text{red}} = \langle C_1 h, C_1 k\rangle_{\mathcal{W}}. \tag{11}$$

The definition $\delta S_{\text{red}} = \langle h, \mathcal{E}(g)\rangle$ and $B_{\text{lin}} = D\mathcal{E}|_{\bar{g}}$ gives $\langle h, B_{\text{lin}}k\rangle = \langle C_1 h, C_1 k\rangle_{\mathcal{W}}$, proving (9). Equation (7) then gives the first detour identity, formal self-adjointness gives the second, and Hessian symmetry gives the third. □

The Weyl-fibre pairing in Lorentz signature is indefinite. Thus $C_1^\sharp C_1$ is a formal-adjoint factorization, not a positivity factorization. In trace-free conformal geometry the corresponding local sequence is

$$\Gamma(T) \xrightarrow{K_0} \Gamma(S_0^2 T^*) \xrightarrow{B_{\mathrm{lin}}} \Gamma(S_0^2 T^*) \xrightarrow{K_0^\sharp} \Gamma(T^*). \tag{12}$$

On Bach-flat backgrounds this is the conformal deformation detour complex of [1]. Formal self-adjointness holds in all signatures; ellipticity is a Riemannian statement and is not assumed on the Lorentzian cylinder.

2.2 The global four-dimensional deformation complex

The detour sequence (12) is not the only complex available on the conformally flat cylinder. On an oriented four-dimensional conformally flat pseudo-Riemannian manifold, the adjoint BGG deformation complex is [2, 5]

$$\Gamma(T) \xrightarrow{K_0} \Gamma(S_0^2 T^*[2]) \xrightarrow{C_1} \Gamma(\mathcal{C}[2]) \xrightarrow{C_1^\sharp \star} \Gamma(S_0^2 T^*[-2]) \xrightarrow{K_0^\sharp} \Gamma(T^*[-2]). \tag{13}$$

Here $\mathcal{C}[2]$ is the algebraic Weyl-curvature bundle and $C_1^\sharp \star$ means $C_1^\sharp \circ \star$, with the Hodge star acting on one antisymmetric index pair. In particular,

$$\boxed{C_1^\sharp \star C_1 = 0.} \tag{14}$$

The repository certificate verifies this identity directly as a constant-coefficient differential-operator identity in both Euclidean and Lorentzian conventions; Gover and Peterson supply the invariant global sequence.

On a locally conformally flat manifold, the flat BGG complex is a fine resolution of the sheaf of parallel sections of the flat adjoint-tractor bundle [6, 5]. The cylinder

$$M = \mathbb{R} \times S^3 \simeq S^3 \tag{15}$$

is simply connected, so that flat local system is the constant system with fibre $\mathfrak{so}(4,2)_{\mathbb{C}}$. Fine-resolution cohomology therefore gives an all-level global statement without an elliptic argument.

Theorem 2.2 (Global smooth deformation cohomology)**.** *For smooth complexified global sections on the conformal cylinder,*

$$H^q_{\mathrm{def}}(M) \cong H^q(S^3; \mathbb{C}) \otimes \mathfrak{so}(4,2)_{\mathbb{C}} \cong \begin{cases} \mathfrak{so}(4,2)_{\mathbb{C}}, & q = 0, 3, \\ 0, & q = 1, 2, 4. \end{cases} \tag{16}$$

In particular, the metric and curvature slots are globally exact:

$$\boxed{\ker C_1 = \operatorname{im} K_0,} \qquad \boxed{\ker(C_1^\sharp \star) = \operatorname{im} C_1.} \tag{17}$$

Proof. The fine resolution computes sheaf cohomology of the flat adjoint local system. Simple connectivity trivializes its monodromy, while $\mathbb{R} \times S^3$ deformation retracts onto S^3. The only nonzero ordinary cohomology groups are in degrees zero and three. Vanishing in degrees one and two is exactly the pair of statements (17). □

The theorem is signature-independent and concerns unrestricted smooth global sections. It does not by itself establish continuity, closed range, or bounded inverses on a Hilbert or Krein completion.

2.3 The Bach quotient as an on-shell curvature space

The passage from deformation-complex exactness to a detour equation is a general exact-sequence argument.

Proposition 2.3 (Detour–curvature lemma)**.** *Let*

$$E_0 \xrightarrow{K} E_1 \xrightarrow{C} E_2 \xrightarrow{D_2} E_3 \tag{18}$$

be a complex which is exact at E_1 and E_2, and let $G : E_2 \to E_1'$ be an operator such that the detour equation is $B = GC$. Then $[h] \mapsto Ch$ induces an isomorphism

$$\boxed{\frac{\ker B}{\operatorname{im} K} \cong \ker D_2 \cap \ker G.} \tag{19}$$

Proof. If $Bh = 0$, then $GCh = 0$, while $D_2Ch = 0$ because the displayed sequence is a complex. If $Ch = 0$, exactness at E_1 gives $h = K\xi$, so the induced map is injective. Conversely, if $U \in \ker D_2 \cap \ker G$, exactness at E_2 gives $U = Ch$; then $Bh = GU = 0$, proving surjectivity. □

The middle local cohomology relevant to the free field is

$$\mathcal{H}_B := \frac{\ker B_{\text{lin}}}{\operatorname{im} K_0}. \tag{20}$$

In the four-dimensional deformation complex,

$$D_2 = C_1^\sharp\star, \qquad G = C_1^\sharp, \qquad B_{\text{lin}} = GC_1. \tag{21}$$

Define the geometrically on-shell curvature space

$$\mathcal{W}_{\text{geom}} := \ker(C_1^\sharp\star) \cap \ker C_1^\sharp. \tag{22}$$

The smooth global BGG theorem and Proposition 2.3 identify the metric quotient with this space exactly.

Theorem 2.4 (Cylinder detour–curvature isomorphism)**.** *For smooth complexified fields on the conformal cylinder, the curvature map $[h] \mapsto C_1h$ induces an $SO(4,2)$-equivariant isomorphism*

$$\boxed{\frac{\ker B_{\text{lin}}}{\operatorname{im} K_0} \cong \mathcal{W}_{\text{geom}} = \ker C_1^\sharp \cap \ker(C_1^\sharp\star).} \tag{23}$$

Proof. If $B_{\text{lin}}h = 0$ and $U = C_1h$, then the factorization (9) gives $C_1^\sharp U = 0$, while (14) gives $C_1^\sharp \star U = 0$. Thus the curvature map lands in the displayed intersection.

If $C_1h = 0$, exactness at the metric slot in (17) gives $h = K_0\xi$, proving injectivity. Conversely, suppose U lies in both kernels. Exactness at the curvature slot gives $U = C_1h$ for a smooth global h, and then $B_{\text{lin}}h = C_1^\sharp U = 0$. Hence every element of the intersection has a Bach-flat metric potential, proving surjectivity. □

In Lorentz signature, complexified Weyl tensors split into Hodge eigenspaces $U = U_+ + U_-$ with $\star U_\pm = \pm iU_\pm$. If $a = C_1^\sharp U_+$ and $b = C_1^\sharp U_-$, the two target equations are

$$a + b = 0, \qquad ia - ib = 0, \tag{24}$$

so $a = b = 0$. Define the smooth on-shell chiral geometric spaces by

$$\mathcal{W}^{\mathrm{sm}}_{\mathrm{geom},\pm} := \{U_\pm : \star U_\pm = \pm i U_\pm,\ C_1^\sharp U_\pm = 0\}. \tag{25}$$

Then Theorem 2.4 becomes

$$\boxed{\frac{\ker B_{\mathrm{lin}}}{\operatorname{im} K_0} \cong \mathcal{W}^{\mathrm{sm}}_{\mathrm{geom},+} \oplus \mathcal{W}^{\mathrm{sm}}_{\mathrm{geom},-}.} \tag{26}$$

All maps commute with $SO(4,2)$. Hence the quotient-to-curvature map preserves compact energy and both rotation spins on quotient classes. It remains to match this geometric system to the abstract positive-energy module constructed in Section 3.

Field-Theoretic Hypothesis 2.5 (Algebraic cylinder realization)**.** Let $\mathcal{W}^{\mathrm{alg},+}_{\mathrm{geom},\pm}$ denote the D-finite, $SO(4)$-finite positive-energy solutions of (25). The curvature construction should restrict to an isomorphism

$$\boxed{\mathcal{W}^{\mathrm{alg},+}_{\mathrm{geom},+} \oplus \mathcal{W}^{\mathrm{alg},+}_{\mathrm{geom},-} \cong \mathcal{W}_+ \oplus \mathcal{W}_-.} \tag{27}$$

Moreover, every element on the left should admit a Bach-flat metric representative which is itself a finite sum in the same D-energy and $SO(4)$ type.

The conjecture has a short, concrete proof target. First construct the curvature intertwiner on every explicit $E/A/L$ oscillator and verify $C_1^\sharp U_\pm = 0$. This gives a graded injection into the geometric kernel. For positive chirality, the flat chiral BGG data consist of the lowest-weight curvature primary $(2;2,0)$, the equation module $(4;1,1)$, and its identity $(5;\frac{1}{2},\frac{1}{2})$. Their alternating character is

$$\chi_{(2;2,0)} - \chi_{(4;1,1)} + \chi_{(5;\frac{1}{2},\frac{1}{2})} = \frac{5q^2 - 9q^4 + 4q^5}{(1-q)^4}. \tag{28}$$

The cylinder module $\mathcal{W}_+$ is cyclic from the same primary and has this character by (34); parity supplies the other chirality. An injective graded intertwiner plus equality of the all-level character would prove the first statement.

The metric-potential statement is separate. The smooth theorem supplies some smooth preimage and equivariance fixes the weight of its *quotient class*; it does not exclude a representative containing secular terms such as te^{-iEt} or an infinite harmonic expansion. One still needs a D- and $SO(4)$-equivariant BGG homotopy, exactness in the relevant Harish–Chandra category, or an explicit all-level harmonic potential. Bounded Poincaré operators illustrate how de Rham homotopies can be transferred to BGG complexes [7], but their existence does not by itself establish this cylinder-mode restriction.

2.4 The finite-jet calculation as an independent convention audit

The earlier finite-jet calculation remains useful, but it is no longer a substitute for an unknown all-level theorem. On homogeneous Euclidean polynomial jets of degree $n = 2, \ldots, 6$, the matrices for K, C_1, and B_1 obey

$$C_1 K = 0, \qquad B_1 K = 0, \qquad \ker C_1 = \operatorname{im} K. \tag{29}$$

The exact ranks are shown in Table 1.

n	$\dim G_n$	$\dim h_n$	$\dim W_{n-2}$	$\dim B_{n-4}$	$\operatorname{rank} K$	$\operatorname{rank} C_1$	$\operatorname{rank} B_1$	$\dim(\ker B_1/\operatorname{im} K)$
2	90	100	10	0	90	10	0	10
3	160	200	40	0	160	40	0	40
4	259	350	100	10	259	91	9	82
5	392	560	200	40	392	168	32	136
6	564	840	350	100	564	276	74	202

Table 1: Exact homogeneous-polynomial detour ranks. The finite Euclidean jet calculation independently audits the conventions and the first five levels of the global BGG theorem.

The quotient dimensions $(10, 40, 82, 136, 202)$ agree with the first five levels of the parity-complete $E/A/L$ module introduced below. The same calculation separates a 15-dimensional kernel of the low-degree gauge map, generated by four translations, six rotations, one dilation, and four special conformal transformations. These finite results are independent regression rails for (17) and (27) in the repository normalization.

2.5 The finite global zero-mode pair

The same topology that kills the metric and curvature cohomology leaves two finite-dimensional groups:

$$H^0_{\mathrm{def}}(M) \cong \mathfrak{so}(4,2)_{\mathbb{C}}, \qquad H^3_{\mathrm{def}}(M) \cong \mathfrak{so}(4,2)_{\mathbb{C}}. \tag{30}$$

The degree-zero copy is the familiar space of fifteen conformal Killing fields. The degree-three copy lies on the adjoint equation side of the self-dual deformation complex. Refining both groups by compact weight gives

$$4_{-1} \oplus 7_0 \oplus 4_{+1}. \tag{31}$$

Poincaré duality on S^3 and the Killing form pair the two copies, with

$$\kappa(\mathfrak{g}_r, \mathfrak{g}_s) = 0 \qquad \text{unless} \qquad r + s = 0. \tag{32}$$

Its dimension, representation, and position are exactly those expected of the dual residual constraint sector. It is therefore a strong candidate for the geometric source of the fifteen moment-map or Taub components constructed in Section 4. That identification is not yet a theorem: the full four-dimensional BGG sequence must first be kept distinct from the shorter Bach detour complex, and the degree-three bundle representative must be located explicitly. Importantly, the direct second-order argument in Proposition 4.1 establishes the Taub constraints without depending on this identification. If the BGG and BFV realizations are then compared, equivariance reduces the map to one nonzero component and one overall scalar.

Thus the smooth deformation complex contains a controlled finite pair, not an unspecified collection of extra global classes. A complete BV argument must still place the degree-three copy at the correct ghost/antifield degree and construct the corresponding zero-mode projector.

2.6 Domain boundary of the theorem

Theorems 2.2 and 2.4 are exact for smooth complexified global sections. Their restriction to the algebraic positive-energy cylinder category remains Conjecture 2.5. The theorems do not automatically identify spacelike-compact or distributional cohomology, nor do they prove continuity and closed range on an infinite Hilbert/Krein completion. If such an analytic completion is required, Green-hyperbolic complexes and causal homotopies remain natural tools [4].

The other remaining issue is BV rather than local geometry. The fifteen conformal-Killing zero modes and their degree-three duals must be split from the local complex without double counting. Dneprov and Grigoriev's minimal nonlinear Weyl-jet model supplies compatible presymplectic data for this comparison [19], but its curvature-dependent ghost terms must not be imported as a proof that the free residual transfer is strict. It does not perform the global cylinder BFV split.

3 The on-shell Weyl module

3.1 Character and cylinder towers

We now define the coefficient module used in the exact residual calculation. A positive-chirality Weyl curvature is a conformal primary with labels

$$(\Delta; j_L, j_R) = (2; 2, 0), \tag{33}$$

and the opposite chirality is its parity conjugate. The Bach equation is in $(4; 1, 1)$ and its differential identity is in $(5; \frac{1}{2}, \frac{1}{2})$. The resulting one-chirality character is

$$\chi_+(q) = \chi_{(2;2,0)} - \chi_{(4;1,1)} + \chi_{(5;\frac{1}{2},\frac{1}{2})} = \frac{5q^2 - 9q^4 + 4q^5}{(1-q)^4}. \tag{34}$$

This defines the abstract algebraic positive-energy module $\mathcal{W}_+$. The character and tower calculation below are the representation-theoretic side of Conjecture 2.5: the smooth metric-to-geometric-curvature map is proved, while equality of the geometric chiral system with this abstract module still requires the explicit intertwiner and algebraic-potential statement described there.

On $\mathbb{R} \times S^3$, $\mathcal{W}_+$ decomposes into three multiplicity-free towers:

$$E_n^+ : \quad \left(\frac{n+2}{2}, \frac{n-2}{2}\right), \qquad \dim E_n = (n+3)(n-1), \qquad n \geq 2, \tag{35}$$

$$A_n^+ : \quad \left(\frac{n}{2}, \frac{n-2}{2}\right), \qquad \dim A_n = (n+1)(n-1), \qquad n \geq 3, \tag{36}$$

$$L_n^+ : \quad \left(\frac{n}{2}, \frac{n-4}{2}\right), \qquad \dim L_n = (n+1)(n-3), \qquad n \geq 4. \tag{37}$$

The negative-chirality towers exchange j_L and j_R. Summing (35)–(37) reproduces (34). For both chiralities the first levels have dimensions

$$\dim \mathcal{H}_2 = 10, \quad \dim \mathcal{H}_3 = 40, \quad \dim \mathcal{H}_4 = 82, \quad \dim \mathcal{H}_5 = 136, \quad \dim \mathcal{H}_6 = 202. \tag{38}$$

The labels E, A, L are cylinder oscillator labels only. In particular, they should not be interpreted as three independently signable conformal representations. Proper conformal transformations mix the towers.

3.2 Invariant form and exact generator action

In normalized cylinder coordinates the unique standard-real, parity-compatible invariant form is, up to overall convention,

$$J_{\text{conf}} = +I_E \oplus (-I_A) \oplus (-I_L). \tag{39}$$

It is nondegenerate and indefinite. The proper conformal lowering and raising generators satisfy

$$(K_q^-)^\sharp = K_q^+, \qquad X^\sharp = X \quad (X = D, SU(2)_L, SU(2)_R), \tag{40}$$

where $X^\sharp = J_{\mathrm{conf}}^{-1} X^\dagger J_{\mathrm{conf}}$.

Multiplicity one reduces each allowed lowering block to a scalar reduced coefficient. With source energy n, the six stable families are

$$\begin{aligned} c_{EE}(n) &= \sqrt{\frac{2(n-1)(n+1)(n+3)}{n+2}}, & c_{AE}(n) &= \sqrt{\frac{8(n-1)}{(n-2)(n+2)}}, \\ c_{AA}(n) &= \sqrt{\frac{2(n-3)(n-1)(n+2)}{n-2}}, & c_{LE}(n) &= -\sqrt{\frac{2(n-3)}{n-2}}, \\ c_{LA}(n) &= \sqrt{\frac{8}{n-2}}, & c_{LL}(n) &= \sqrt{2(n-2)(n+1)}. \end{aligned} \tag{41}$$

Together with Clebsch–Gordan coefficients and (40), these determine the all-level one-particle $\mathfrak{so}(4,2)$ action. The exact reconstruction verifies the conformal commutators on every interior level of a finite energy buffer; the top shell is used only as a buffer and is never mistaken for a finite conformal representation.

The free Fock coefficient module is the bosonic second quantization

$$\mathcal{F}_{\mathcal{W}} = \mathrm{Sym}(\mathcal{W}_+ \oplus \mathcal{W}_-), \tag{42}$$

with multiplicative lift of (39). The residual generators preserve oscillator particle number.

4 Hamiltonian moment-map normalization

The residual action is not introduced as an arbitrary matrix representation. In oscillator coordinates z, write the symplectic form as

$$\Omega = i\, d\bar{z}\, J \wedge dz. \tag{43}$$

If a conformal generator X acts linearly by $\delta_X z = K_X z$, its quadratic Hamiltonian moment map is

$$\mu_X(z,\bar{z}) = \bar{z}\, M_X z, \qquad M_X = J K_X, \qquad d\mu_X = \iota_{X_X}\Omega. \tag{44}$$

The J-adjoint identities ensure the correct reality of μ_X.

4.1 The conformal Taub constraint

The same residual parameters arise as reducibilities of the local Diff×Weyl complex. Let

$$\epsilon = (\xi, \sigma), \qquad K(\xi,\sigma) = 0, \qquad \sigma = -\frac{1}{4}\bar{\nabla}_\mu \xi^\mu, \tag{45}$$

and let h satisfy $B^{(1)}[h] = 0$. With the Taylor convention

$$g(\lambda) = \bar{g} + \lambda h + \frac{\lambda^2}{2} k + O(\lambda^3), \tag{46}$$

the second-order equation is

$$B^{(1)}[k] + B^{(2)}[h,h] = 0. \tag{47}$$

Proposition 4.1 (Conformal Taub solvability constraint)**.** *For every residual conformal reducibility ϵ, the current*

$$J^\mu_\epsilon[h] := \xi_\nu B^{(2)\mu\nu}[h,h] \tag{48}$$

is conserved. If h is tangent to a second-order family of Bach-flat metrics on the compact cylinder, then

$$\boxed{\mathcal{T}_\epsilon[h] := \int_{S^3} n_\mu \xi_\nu B^{(2)\mu\nu}[h,h], d\Sigma = 0.} \tag{49}$$

Proof. At second order around a Bach-flat background, the exact divergence and trace identities imply $\bar{\nabla}_\mu B^{(2)\mu\nu}[h,h] = 0$ and $B^{(2)\mu}{}_\mu[h,h] = 0$ whenever $B^{(1)}[h] = 0$. Hence

$$\bar{\nabla}_\mu J^\mu_\epsilon = B^{(2)\mu\nu}\bar{\nabla}_{(\mu}\xi_{\nu)} = -\sigma B^{(2)\mu}{}_\mu = 0. \tag{50}$$

If (47) has a solution, the charge equals minus the integral of $\xi_\nu B^{(1)\mu\nu}[k]$. For a reducibility parameter, the linearized Noether current is a spatial divergence. Since $\partial S^3 = \varnothing$, its integral vanishes. This is the standard compact-Cauchy linearization-stability argument [21, 22]. □

The covariant phase-space Noether identity predicts that this same quadratic constraint is the Hamiltonian of the residual linear action:

$$\mu_\epsilon(h) = \frac{1}{2}\Omega_\Sigma(h, R_\epsilon h) \stackrel{?}{=} \pm\mathcal{T}_\epsilon[h]. \tag{51}$$

The sign and a possible factor of two depend only on the Taylor convention for $B^{(2)}$. Both sides form equivariant adjoint/coadjoint $\mathfrak{so}(4,2)$ modules. Since the complexified adjoint module is simple, an equivariant nonzero comparison is unique up to one scalar. Consequently the all-component equality can be proved by establishing equivariance and matching one generator, preferably D. This is also the local-BRST relation between reducibilities and conserved-current classes [16, 15].

In the curved-cylinder wave normalization used for the independent Taub calculation, comparison with the quadratic action fixes the proper-conformal kernel to

$$M_{\text{Taub}} = -\frac{\sqrt{2}}{4\pi}\, JK^- \tag{52}$$

up to the explicitly declared orientation and harmonic phases. Two independent curvature components fix the common coefficient; it is not a one-entry fit. An exact two-chirality jet through source energy four verifies all fifteen moment-map components and the full conformal algebra on interior energies two and three.

Equation (52) verifies (51) for the tested proper-conformal kernels. The remaining all-level task is the single equivariant normalization comparison, not fifteen unrelated curvature calculations.

This section supplies the Hamiltonian normalization of the residual action and its Taub interpretation. The cohomology theorem below depends only on the exact module action, not on interpreting a finite moment-map jet as a physical quotient.

5 Residual BRST and Cartan homotopy reduction

5.1 The absolute residual complex

Let

$$\mathfrak{g} = \mathfrak{so}(4,2) \tag{53}$$

with compact grading

$$\mathfrak{g} = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_{+1}, \qquad \dim(\mathfrak{g}_{-1}, \mathfrak{g}_0, \mathfrak{g}_{+1}) = (4,7,4). \tag{54}$$

Here D and $\mathfrak{so}(4)$ lie in $\mathfrak{g}_0$, while proper conformal lowering and raising generators lie in $\mathfrak{g}_{-1}$ and $\mathfrak{g}_{+1}$. The dual ghosts carry the opposite compact degree.

For a coefficient module $V \subset \mathcal{F}_{\mathcal{W}}$, the absolute residual complex is

$$C^q(V) = V \otimes \Lambda^q \mathfrak{g}^*, \tag{55}$$

with Chevalley–Eilenberg differential

$$d = c^a \rho(G_a) - \frac{1}{2} f^a{}_{bc} c^b c^c \iota_a. \tag{56}$$

It raises ghost number by one and preserves the total compact degree

$$\delta = D_{\text{matter}} + D_{\text{ghost}}. \tag{57}$$

The standard centered polarization uses the product v of the four ghosts dual to the raising generators. It has

$$\text{gh}(v) = 4, \qquad D_{\text{ghost}}(v) = -4. \tag{58}$$

This is the residual ghost convention used in the cylinder construction of [11]. Its compact degree and uniqueness as the bottom rotational scalar follow from the $(4,7,4)$ exterior-algebra polarization, and Lemma 6.1 shows that its complementary-degree pairing is canonical up to one overall volume. Only the match of that scalar to the strict-pure-Weyl BV/BFV normalization remains a hypothesis in Section 8; the broader Riegert sector of [11] is not included.

5.2 Full residual gauging is a physical assumption

Hamada's background-independence programme identifies the transformations which preserve the cylinder gauge with residual diffeomorphisms and imposes the resulting conformal charges through BRST [10, 11]. This supplies a direct precedent for the complex (55). It does not prove that every quantization of strict pure-Weyl gravity must use the same quotient.

There are at least two distinct cylinder problems.

1. In the *fully gauged residual problem*, the cylinder is a gauge representative of a closed generally covariant system. All fifteen conformal-Killing transformations, including compact time translation, are generated by residual constraints. The ghost for D exists and the Cartan homotopy below is an allowed operation.

2. In the *fixed-background cylinder problem*, D is the Hamiltonian and $SO(4,2)$ is a global symmetry acting on states or observables. Its eigenspaces are not quotiented. The D ghost and its contraction cannot be used to erase nonzero energy.

Proposition 4.1 gives the first choice a direct linearization-stability interpretation. Within the closed-universe Dirac boundary problem, the slice S^3 has no spatial boundary supporting an independent asymptotic charge, and the residual conformal Killing parameters instead generate quadratic integrability constraints, including the D constraint. Subject to the all-level normalization in (51), full residual gauging is therefore a coherent zero-boundary-charge sector of pure conformal gravity; it is not a device introduced merely to erase positive cylinder energy. Compactness and the Taub identity support this interpretation but do not make it the unique physical quantization.

Adding a boundary or asymptotic region, coupling an external clock, or deparametrizing the cylinder can convert D into a physical global charge. That is the second problem above, not a contradiction of the first. Because $[K^-, K^+]$ contains D, the second problem is not obtained by simply deleting one ghost while leaving the rest of (56) unchanged. It requires a different relative or boundary-charge complex. Theorem 6.2 answers only the first problem. A strict pure-Weyl derivation must state the temporal boundary conditions, the residual gauge group, and the treatment of its charges before calling the result a physical state space.

5.3 Cartan homotopy

Let ι_D be contraction with the generator D in the residual ghost exterior algebra. The usual Cartan identity is

$$d\iota_D + \iota_D d = \mathcal{L}_D. \tag{59}$$

Theorem 5.1 (Cartan homotopy reduction to compact degree zero)**.** *Decompose the residual complex into total compact-degree eigenspaces,*

$$C = \bigoplus_{\delta} C_\delta, \qquad \mathcal{L}_D|_{C_\delta} = \delta \,\mathrm{id}. \tag{60}$$

Every C_δ with $\delta \neq 0$ is contractible. Hence inclusion of the centered subcomplex is a quasi-isomorphism:

$$\boxed{H^\bullet(C, d) \cong H^\bullet(C_{\delta=0}, d).} \tag{61}$$

Proof. On C_δ with $\delta \neq 0$, define

$$h_\delta = \frac{\iota_D}{\delta}. \tag{62}$$

Equation (59) gives

$$dh_\delta + h_\delta d = \mathrm{id}. \tag{63}$$

Thus every closed vector of nonzero total degree has the explicit primitive $\delta^{-1}\iota_D\Psi$. If $\mathcal{L}_D$ has a nilpotent part, the same proof works wherever $\mathcal{L}_D$ is invertible by using $h = \iota_D\mathcal{L}_D^{-1}$. □

The theorem depends on treating D as a residual gauge generator. If cylinder time translation is retained as a physical Hamiltonian or a boundary charge, contraction by its ghost is not an allowed quotient and (61) does not apply.

5.4 Finite centered inventory

At ghost number four, the minimal residual exterior algebra contains at most the four negative-degree ghosts, so

$$D_{\mathrm{ghost}} \geq -4. \tag{64}$$

At $\delta = 0$ this implies

$$D_{\text{matter}} \leq 4. \tag{65}$$

Every nonvacuum Weyl quantum has weight at least two. Therefore only particle numbers $N = 0, 1, 2$ can contribute to the centered ghost degree selected by the Hamada polarization. All $N \geq 3$ sectors have matter weight at least six and are contractible before any matrix-rank computation.

6 Complete centered cohomology

6.1 The relative weight-four kernel

The complete matter Fock space at weight four is

$$\mathcal{F}_4 = \mathcal{H}_4 \oplus \operatorname{Sym}^2 \mathcal{H}_2, \qquad \dim \mathcal{F}_4 = 82 + 55 = 137. \tag{66}$$

The induced matter form has signature

$$\operatorname{sig}(J_{\mathcal{F}_4}) = (97, 40). \tag{67}$$

The relative conformal-primary conditions are

$$(D - 4)|\Psi\rangle = 0, \qquad R|\Psi\rangle = 0, \qquad K_q^-|\Psi\rangle = 0. \tag{68}$$

Their exact kernel on the full space (66) is two-dimensional. In a spin-two magnetic basis its normalized chiral generators are

$$|W_\pm^2\rangle = \sqrt{\frac{2}{5}}|2, -2\rangle_\pm - \sqrt{\frac{2}{5}}|1, -1\rangle_\pm + \frac{1}{\sqrt{5}}|0, 0\rangle_\pm. \tag{69}$$

They obey

$$\langle W_r^2 | J_{\mathcal{F}} | W_s^2 \rangle = \delta_{rs}. \tag{70}$$

This relative computation is not by itself an absolute cohomology theorem; incoming exact states must also be excluded.

There is a simple representation-theoretic reason for the number two. The weight-two one-particle space is

$$E_2^+ \oplus E_2^- = (2, 0) \oplus (0, 2). \tag{71}$$

The symmetric square decomposes as

$$\operatorname{Sym}^2(2, 0) \oplus \big[(2, 0) \otimes (0, 2)\big] \oplus \operatorname{Sym}^2(0, 2). \tag{72}$$

There is one compact scalar in each same-chirality symmetric square and no scalar in the mixed product. Since $E_2^\pm$ are lowest-weight spaces, every K_q^- annihilates both factors, so these two scalar singlets are relative primaries. By contrast, the one-particle weight-four space contains

$$(3, 1) \oplus (2, 1) \oplus (2, 0) \tag{73}$$

and its parity conjugate, with no compact scalar. Thus representation theory predicts the two relative candidates before the matrix calculation. The absolute computation below is still necessary to exclude exactness and to prove the one-particle acyclicity. A complete Hochschild–Serre or Kostant-style derivation of the latter ranks would be a useful conceptual replacement for the large matrices [5].

6.2 Absolute kernel and image

The cutoff-complete absolute calculation supplies that missing test. For one chirality in the one-particle coefficient module, the centered complex around ghost number four has exact dimensions

$$C_0^3(290) \xrightarrow{d_3} C_0^4(1311) \xrightarrow{d_4} C_0^5(3657). \tag{74}$$

The entries lie in $\mathbb{Q}(i, \sqrt{2}, \sqrt{3}, \sqrt{5})$. Good-prime reduction modulo 241, combined with exact nilpotency, fixes the characteristic-zero ranks to

$$\operatorname{rank} d_3 = 260, \qquad \operatorname{rank} d_4 = 1051. \tag{75}$$

Since the ranks sum to 1311,

$$H^4_{\delta=0,N=1} = 0. \tag{76}$$

Parity supplies the identical result for the opposite chirality.

The lowest two-particle block begins at matter weight four. Exterior saturation makes its beginning

$$0 \longrightarrow C_0^4(55) \xrightarrow{d_4} C_0^5(385) \xrightarrow{d_5} C_0^6(1155). \tag{77}$$

There is no incoming C_0^3 space: three residual ghosts have compact degree at least -3 and cannot center a two-particle coefficient of minimum weight four. Exact nilpotency gives $d_5 d_4 = 0$. The two vectors (69) lie in $\ker d_4$, while a good-prime minor gives $\operatorname{rank} d_4 \geq 53$. The displayed independent kernel vectors give $\operatorname{rank} d_4 \leq 53$. Therefore

$$H^4_{\delta=0,N=2} = \operatorname{span}\{[W_+^2], [W_-^2]\}. \tag{78}$$

For the vacuum coefficient module the complex is ordinary Lie-algebra cohomology. Since

$$H^\bullet(\mathfrak{so}(4,2); \mathbb{C}) \cong \Lambda(u_3, u_5, u_7), \tag{79}$$

one has

$$H^4_{\delta=0,N=0} = 0. \tag{80}$$

6.3 The canonical residual CE pairing

The residual ghost normalization is fixed almost completely by the compact grading. Write

$$\mathfrak{g} = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_{+1}, \qquad \dim(\mathfrak{g}_{-1}, \mathfrak{g}_0, \mathfrak{g}_{+1}) = (4, 7, 4), \tag{81}$$

and choose nonzero determinant vectors

$$v_- \in \det(\mathfrak{g}_{+1}^*), \qquad \Theta_0 \in \det(\mathfrak{g}_0^*), \qquad v_+ \in \det(\mathfrak{g}_{-1}^*), \tag{82}$$

with $v_+ = v_-^\dagger$. Their product

$$\Omega_\mathfrak{g} = v_+ \wedge \Theta_0 \wedge v_- \tag{83}$$

is the top Berezin volume on all fifteen residual ghosts. Define the polarized pairing by coefficient extraction,

$$(\alpha, \beta)_{\text{gh}} := [\alpha^\dagger \wedge \Theta_0 \wedge \beta]_{\Omega_\mathfrak{g}}. \tag{84}$$

Lemma 6.1 (Canonical centered ghost pairing)**.** *After fixing the orientation and the overall residual volume,* (84) *is the unique determinant-saturating pairing compatible with the* $(4,7,4)$ *polarization. It obeys*

$$(v_-, v_-)_{\mathrm{gh}} = 1. \tag{85}$$

Because $\mathfrak{so}(4,2)$ *is semisimple and hence unimodular, the Chevalley–Eilenberg differential is graded skew for the associated complementary-degree Poincaré pairing.*

Proof. Each of the three determinant lines is one-dimensional. Once their product is oriented and normalized, coefficient extraction gives (85) and leaves no matrix freedom. Unimodularity makes the top CE volume invariant, so integration by parts in the exterior algebra gives graded skew-adjointness of the CE differential. □

Consequently exact cochains are orthogonal to closed complementary-degree cochains, and the pairing descends to residual CE cohomology.

In Hamada's oscillator notation, v_- is the product of the four proper-conformal ghosts, Θ_0 contains the dilation/time ghost and the six rotation ghosts, and the bra supplies the four conjugate ghosts [11]. The factor of i is a real-structure convention. The repository exterior-algebra certificate verifies the saturation, Hermiticity, compact degree -4, and unit overlap directly.

The fixed insertion is not a nondegenerate inner product on every individual ghost-number block. It is the polarized form of the canonical complementary-degree CE pairing. It is sufficient here because the incoming two-particle space is empty and the two normalized representatives are fixed. Multiplying (85) by the matter restriction (70) gives their exact residual cohomological Gram matrix. It is Hermitian and sesquilinear on the complex oscillator module; the parity-compatible real structure gives its real restriction. This is an intrinsic statement about the chosen residual CE complex. We reserve the phrase "transferred pure-Weyl BV cohomological pairing" for the cyclic-transfer statement in Section 8. Showing that the full local BV/BFV reduction induces this residual orientation and overall volume is part of the conditional bridge.

Theorem 6.2 (Complete cohomology under full residual conformal gauging)**.** *Let the coefficient module be the free bosonic Fock space* $\mathcal{F}(\mathcal{W}_+ \oplus \mathcal{W}_-)$*, let the residual algebra be* $\mathfrak{so}(4,2)$*, treat* D *as a gauge generator, and use the centered ghost polarization* (58)*. Then the complete minimal absolute residual cohomology at the selected ghost number four is*

$$\boxed{H^4_{\mathrm{res,min}} = \mathrm{span}\{[W_+^2], [W_-^2]\}, \qquad G_{\mathrm{res}} = I_2.} \tag{86}$$

There is no one-particle class and no higher-matter-weight class in this minimal centered polarization.

Proof. The Cartan homotopy, Theorem 5.1, eliminates every nonzero total compact degree. The ghost-degree floor then reduces the centered inventory to $N = 0, 1, 2$. Equations (80), (76), and (78) exhaust those sectors. Equations (70) and (85) give the pairing. □

Remark 6.3 (What is positive). The theorem does not turn the indefinite one-particle module (39) into a positive particle space. It removes the one-particle sector from this absolute residual cohomology. The positive I_2 belongs to two ghost-dressed scalar vertex classes.

7 Descent and the dynamical/topological split

7.1 Why weight four is selected

Let V_h be a scalar conformal primary of weight h, and let

$$\omega = \frac{1}{4!}\epsilon_{\mu\nu\rho\sigma}c^\mu c^\nu c^\rho c^\sigma \tag{87}$$

be the top lowering-ghost product. In the residual descent conventions,

$$sV_h = c^\mu \nabla_\mu V_h + \frac{h}{4}(\nabla_\mu c^\mu)V_h, \qquad s\omega = -\omega \nabla_\mu c^\mu, \qquad \omega c^\mu = 0. \tag{88}$$

It follows that both the integrated density and the unintegrated ghost-dressed operator have the same closure defect, proportional to $h/4 - 1$. Hence

$$\boxed{[\omega V_4] \longleftrightarrow \left[\int_M d^4x\,\sqrt{-g}\,V_4\right].} \tag{89}$$

The -4 compact degree of the ghost factor is therefore the conformal counterpart of the four-dimensional volume form, rather than an accidental normal-ordering shift. This weight-four state–operator mechanism and a Weyl-square example were already explicit in [11]; the new use here is the absolute exhaustion and pairing calculation of Section 6.

Three cohomological quotients appear and must not be identified without the transfer theorem:

$$H^4\big(\mathfrak{so}(4,2);\mathcal{F}_{\mathcal{W}}\big), \qquad H^{0,4}_{\rm loc}(s \mid d), \qquad \frac{\{\text{integrated local functionals}\}}{\{\text{allowed boundary functionals}\}}. \tag{90}$$

The first is the residual Chevalley–Eilenberg group computed exactly in this paper. The second is local BRST cohomology modulo spacetime descent [15]. The third depends on topology and boundary conditions. Equation (89) identifies representatives in the residual construction; its promotion to an isomorphism with the strict pure-Weyl local group is part of the conditional bridge.

7.2 Parity basis

Parity exchanges W_+^2 and W_-^2. Define

$$e = \frac{W_+^2 + W_-^2}{\sqrt{2}}, \qquad o = \frac{W_+^2 - W_-^2}{\sqrt{2}}. \tag{91}$$

Up to the orientation-dependent factor of i in Lorentzian self-duality conventions, descent identifies

$$e \sim C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}, \qquad o \sim C_{\mu\nu\rho\sigma}\widetilde{C}^{\mu\nu\rho\sigma}. \tag{92}$$

Because (91) is orthogonal, its Gram matrix remains I_2.

7.3 Euler–Lagrange quotient

Let

$$\mathcal{E} : H^4_{\mathrm{res,min}} \longrightarrow \{\text{local equation and gauge deformations}\} \tag{93}$$

be the Euler–Lagrange map obtained by varying the integrated density. In the conventions of Section 2,

$$\mathcal{E}(e) = \lambda_B B_{\mu\nu}, \qquad \lambda_B \neq 0, \qquad \mathcal{E}(o) = 0. \tag{94}$$

The first identity is the Weyl-square variation. Under locality, smoothness, Lorentz invariance, and a four-derivative bound, it is the unique nontrivial self-interaction of a single linear conformal graviton modulo equivalence [14].

The second identity is Chern–Weil transgression. In a local frame,

$$P_4 = \mathrm{Tr}(R \wedge R) = dQ_3(\Gamma), \tag{95}$$

where

$$Q_3(\Gamma) = \mathrm{Tr}\left(\Gamma \wedge d\Gamma + \frac{2}{3}\Gamma \wedge \Gamma \wedge \Gamma\right). \tag{96}$$

Its variation is

$$\delta P_4 = 2d\,\mathrm{Tr}(\delta\Gamma \wedge R), \tag{97}$$

using the Bianchi identity. On a finite cylinder, in a chosen global frame,

$$\int_{[t_1,t_2]\times S^3} P_4 = \mathrm{CS}[\Gamma(t_2)] - \mathrm{CS}[\Gamma(t_1)]. \tag{98}$$

Thus fixed-endpoint or compactly supported variations have no bulk Euler–Lagrange derivative.

Equations (94)–(98) give the exact sequence

$$0 \longrightarrow \mathrm{span}\{o\} \longrightarrow H^4_{\mathrm{res,min}} \xrightarrow{\mathcal{E}} \mathrm{span}\{B_{\mu\nu}\} \longrightarrow 0. \tag{99}$$

Quotienting variationally trivial local bulk functionals yields

$$H^4_{\mathrm{dyn}} = H^4_{\mathrm{res,min}}/\ker\mathcal{E} \cong \mathrm{span}\{[C^2]\}, \qquad \boxed{G_{\mathrm{dyn,res}} = I_1.} \tag{100}$$

Equivalently,

$$\boxed{I_2 = I_1^{\mathrm{dynamical}} \oplus I_1^{\mathrm{topological}}.} \tag{101}$$

The odd class is only locally and variationally trivial. A gravitational theta parameter can affect boundaries, horizons, contact terms, large frame transformations, and nontrivial topology [24]. We do not quotient those global data without specifying the boundary problem.

8 The local-to-residual transfer

Theorem 6.2 begins with the locally reduced Weyl module. This section states exactly what is needed to identify it with the complete free pure-Weyl BV answer.

8.1 The free Fock lift is formal

Proposition 8.1 (Algebraic symmetric-algebra lift)**.** *Let (C, q) be the algebraic one-particle free complex over characteristic zero, with q extended as a derivation to* $\operatorname{Sym} C$*. Then*

$$\boxed{H(\operatorname{Sym} C, q) \cong \operatorname{Sym} H(C, q).} \tag{102}$$

In particular, once the one-particle coefficient module is $\mathcal{W}_+ \oplus \mathcal{W}_-$, its free many-particle cohomology is $\mathcal{F}(\mathcal{W}_+ \oplus \mathcal{W}_-)$.

Proof. Degree by degree, $C^{\otimes N}$ obeys the Künneth theorem. Taking symmetric coinvariants under the finite group S_N is exact in characteristic zero, so $H(\operatorname{Sym}^N C, q) \cong \operatorname{Sym}^N H(C, q)$. Taking the algebraic direct sum over N proves (102). □

Thus the Fock-space coefficient module is not an independent field-theory hypothesis. The statement is algebraic; topological tensor products and completed Fock spaces require their own continuity assumptions.

8.2 Equivariant Cartan transfer

Let (C, q) be the gauge-fixed local free BV complex after the fifteen conformal-Killing zero modes have been separated, and let $H = H(q)$. Suppose there is a strong deformation retract

$$(H, 0) \underset{p}{\overset{\jmath}{\rightleftarrows}} (C, q), \qquad \jmath p = \mathrm{id} - qs - sq, \qquad p\jmath = \mathrm{id}_H. \tag{103}$$

Let Δ contain the residual zero-mode differential and mixed terms, so $Q = q + \Delta$. With the sign convention in (103), homological perturbation gives

$$I = (\mathrm{id} + s\Delta)^{-1}\jmath, \qquad P = p(\mathrm{id} + \Delta s)^{-1}, \qquad Q_H = p\Delta(\mathrm{id} + s\Delta)^{-1}\jmath. \tag{104}$$

If

$$[D, q] = [D, \Delta] = [D, s] = 0, \qquad pD = D_H p, \qquad D\jmath = \jmath D_H, \tag{105}$$

then the transferred operators

$$\iota_D^H = P\iota_D I, \qquad \mathcal{L}_D^H = P\mathcal{L}_D I \tag{106}$$

obey

$$[Q_H, \iota_D^H]_+ = \mathcal{L}_D^H. \tag{107}$$

Thus every transferred subspace on which $\mathcal{L}_D^H$ is invertible is again contractible.

8.3 Cyclic transfer and the pairing

Let $\sharp$ be the graded adjoint for the full BV/Krein pairing. Suppose the unperturbed retract is cyclic,

$$\jmath^\sharp = p, \qquad (s\Delta)^\sharp = \Delta s. \tag{108}$$

Then taking the adjoint of (104) gives

$$I^\sharp = P. \tag{109}$$

Since $PI = \mathrm{id}_H$,

$$\boxed{I^\sharp I = \mathrm{id}_H.} \tag{110}$$

The dressed inclusion is therefore an exact isometry: contractible local dressing changes representatives without changing the reduced Gram matrix.

Proposition 8.2 (Blockwise cyclic retract)**.** *Let a finite compact-energy and $SO(4)$ block carry a nondegenerate graded pairing, let $q^\sharp = -q$, and suppose the induced pairing on $H(q)$ is nondegenerate. Then the block admits a cyclic strong deformation retract. The algebraic direct sum of these retracts may be chosen compact-degree and $SO(4)$ equivariant.*

Proof. Write $B = \operatorname{im} q$ and choose a nondegenerate representative space $\mathcal{H} \subset \ker q$. In $\mathcal{H}^\perp$, choose an isotropic complement L to B. Then $q : L \to B$ is an isomorphism. Define $s|_B = (q|_L)^{-1}$ and set $s = 0$ on $\mathcal{H} \oplus L$. The decomposition $C = \mathcal{H} \oplus B \oplus L$ gives $qs + sq = 1 - \jmath p$, while the hyperbolic pairing of B and L gives the cyclic adjoint relation. Choosing the splitting in each finite symmetry block and taking the algebraic direct sum preserves compact degree and $SO(4)$ type. □

This proposition does not provide one splitting which is simultaneously equivariant under the noncompact generators $K^\pm$. Such a splitting cannot be obtained by averaging over the nonunitary conformal module, which need not be semisimple. Compatibility with the full $SO(4,2)$ action, the global zero-mode split, and the BV pairing remains a field-theory obligation. Boundedness on a completed cylinder space is a separate analytic question.

8.4 The filtered local-to-residual complex

The possible failure modes can be displayed without compressing them into the phrase "no other row." After the conformal-Killing zero modes have been split from the local ghosts, introduce the residual-ghost filtration on the bicomplex

$$C^{p,q} = \Lambda^p \mathfrak{g}^* \widehat{\otimes} C^q_{\text{local}}, \qquad F^p C = \bigoplus_{r \geq p} C^{r,*}. \tag{111}$$

Here p is residual ghost number and q is local BV degree. A transferred degree-one differential has the filtered expansion

$$Q_H = q_{\text{local}} + d_{\text{res}} + \sum_{k \geq 2} Q_{[k]}, \tag{112}$$

with bidegrees

$$|q_{\text{local}}| = (0,1), \qquad |d_{\text{res}}| = (1,0), \qquad |Q_{[k]}| = (k, 1-k). \tag{113}$$

The terms $Q_{[k]}$ encode higher transferred ghost operations; assuming that they vanish is a substantive strictness assertion, not automatic homological bookkeeping.

The resulting spectral sequence begins

$$E_1^{p,q} = \Lambda^p \mathfrak{g}^* \otimes H^q(q_{\text{local}}), \qquad E_2^{p,q} = H^p(\mathfrak{g}; H^q(q_{\text{local}})), \tag{114}$$

and

$$d_r : E_r^{p,q} \longrightarrow E_r^{p+r,q-r+1}. \tag{115}$$

For the desired term $E_r^{4,0}$, a higher differential can enter from $(4-r, r-1)$ for $r = 2,3,4$, and can leave toward $(4+r, 1-r)$ for $2 \leq r \leq 11$. Thus local classes in degrees $q = 1,2,3$ and antifield or negative-degree classes down to $q = -10$ are potential modifiers. They must be inventoried or excluded; it is not enough to compute only the local degree-zero row. Since $0 \leq p \leq 15$, this filtration is bounded. On each finite compact-energy block the cochain spaces used here are finite, so the spectral sequence converges. An infinite completed state space additionally requires a complete, separated filtration. Even after collapse, a cyclic splitting is needed to control extension data and the Hermitian form.

This is the standard local-BRST setting in which antifield and gauge-fixed cohomologies must be compared carefully [15, 17].

Proposition 8.3 (Single-row strictness and collapse)**.** *Suppose the global zero modes have been split exactly, $H^q(q_{\mathrm{local}}) = 0$ for $q \neq 0$, and the free residual perturbation Δ has bidegree $(1, 0)$. Suppose in addition that the chosen SDR is compatible with the local grading, so that s has bidegree $(0, -1)$ and $\jmath, p$ are supported in local degree zero. Then*

$$\boxed{Q_H = p\Delta\jmath,} \tag{116}$$

all higher HPL terms vanish, and the local-to-residual spectral sequence collapses to

$$H^p\big(\mathfrak{g}; H^0(q_{\mathrm{local}})\big)\,. \tag{117}$$

Proof. The homotopy s lowers local degree by one. Since both $\jmath(H)$ and the image retained by p lie in local degree zero,

$$p\Delta(s\Delta)^r\jmath = 0 \qquad (r \geq 1). \tag{118}$$

The HPL series therefore reduces to (116). The E_1 page has only the row $q = 0$, so no higher differential can enter or leave it. □

The proposition concerns the perturbation obtained from the *free metric BV bicomplex.* It must not be inferred by naively restricting a nonlinear minimal Q-manifold. In the Dneprov–Grigoriev variables, curvature-dependent terms schematically of the form $\xi\xi W$ and $\xi\xi C$ occur in ghost transformations and can have total compact degree zero [19]. They belong to the nonlinear comparison and must either be identified with the interacting deformation or removed by the equivalence back to the free metric presentation before Proposition 8.3 is applied.

Proposition 8.4 (Formal filtered-transfer criterion)**.** *Suppose the zero-mode split produces (111); the filtration is convergent; $H^q(q_{\mathrm{local}})$ vanishes in every row that can enter or leave $(4, 0)$; and the transferred differential on $H^0(q_{\mathrm{local}})$ is the strict residual Chevalley–Eilenberg differential, with all relevant $Q_{[k]}$ absent. Then the full cohomology in the selected filtered degree is isomorphic, as a vector space, to the residual group*

$$H^4\big(\mathfrak{g}; H^0(q_{\mathrm{local}})\big)\,. \tag{119}$$

If in addition the retract is compact-degree equivariant and cyclic, the Cartan homotopy and the cohomological Hermitian pairing transfer, and the dressed inclusion is the isometry (110).

Proof. Under the stated row and strictness assumptions no differential (115) enters or leaves $(4, 0)$, so the bounded spectral sequence collapses there to its E_2 value. The equivariant and cyclic statements are (107) and (110). The cyclic splitting fixes the extension of the pairing from the associated graded object. □

8.5 Three remaining field-theoretic hypotheses

The smooth BGG theorem removes the original global PDE-surjectivity gap, but it does not by itself compute the complete algebraic BV complex. The remaining bridge is concentrated in three statements.

Field-Theoretic Hypothesis 8.5 (Strict pure-Weyl transfer hypotheses)**.** The gauge-fixed free strict-pure-Weyl BV complex has the following properties in the cylinder problem under consideration.

1. *Algebraic cylinder realization.* Conjecture 2.5 holds: the geometric chiral solution spaces and their Bach-flat metric potentials restrict exactly to the D-finite, $SO(4)$-finite $E/A/L$ modules used in the residual calculation.

2. *BV transfer.* The fifteen global reducibilities and their genuine dual constraint/antifield sectors can be split without double counting. After nonminimal doublets are removed, the remaining nonzero-mode local BV complex admits an $SO(4,2)$-equivariant cyclic retract to $\mathcal{F}(\mathcal{W}_+ \oplus \mathcal{W}_-)$, has no additional local or antifield row which can enter or leave the centered degree, and transfers the strict linear residual action with no higher ghost operations.

3. *Residual BFV choice.* The selected closed-universe boundary problem gauges all fifteen residual generators and its BFV pairing induces the compact-energy four-ghost polarization and complementary-degree determinant insertion of Lemma 6.1, including the common normalization (85) and the parity-compatible real structure.

These are not consequences of the residual rank matrices. The first can be closed by an explicit cylinder intertwiner and equivariant potential construction. The second requires the complete local ghost/antifield row inventory and a genuinely conformal cyclic transfer, rather than merely the blockwise $D \times SO(4)$ splitting of Proposition 8.2. The third is stronger than verifying the natural Berezin saturation inside the already chosen residual complex.

Corollary 8.6 (Conditional algebraic strict-pure-Weyl result)**.** *If Field-Theoretic Hypothesis 8.5 holds, the complete free classical local-plus-residual BV cohomology in the algebraic cylinder category and selected degree is* (86)*. Its transferred cohomological Hermitian form is I_2, and its local Euler–Lagrange quotient is* (100) *with form I_1.*

Proof. Item 1 and Theorem 2.4 identify the one-particle coefficient module. Proposition 8.1 gives its free Fock lift. Item 2 supplies the full equivariant cyclic hypotheses under which Propositions 8.2 and 8.3 give strict residual transfer and spectral-sequence collapse. Item 3 fixes the selected ghost pairing and normalization. The exact residual theorem then gives (86), while (110) identifies its residual CE Gram matrix with the transferred pure-Weyl BV cohomological form. □

The global smooth BGG theorem, curvature isomorphism (23), action factorization (9), and finite jet table are exact inputs. Separate finite-dimensional fixtures verify (107) and (110) with a genuinely dressed inclusion. They do not instantiate the full pure-Weyl field complex. Nor does the smooth BGG preimage prove exactness in the algebraic cylinder category. The three hypotheses above state the remaining obligations without folding algebraic mode exactness, full conformal equivariance, antifield-row control, or BFV normalization into the smooth theorem. A Hilbert, Krein, distributional, or completed Fock statement would require separate continuity, closed-range, and convergence results.

9 Scope, interpretation, and open questions

Residual gauge choice. The result is specific to quotienting the full residual $SO(4,2)$ action on the compact cylinder. If some generators are retained as global or asymptotic charges, the physical complex is different. In particular, Cartan contraction cannot be used to erase nonzero D energy when D is the physical Hamiltonian.

No positive graviton Fock claim. The unreduced module remains indefinite and the one-particle residual cohomology is zero. The positive classes are weight-four scalar vertices. Calling I_2 a positive graviton Hilbert space would therefore misidentify both the cohomological degree and the state–operator map.

Classical and free. The result concerns the free classical complex and its finite residual reduction. It does not show that the interaction preserves the reduced space or pairing. Nor does it address the quantum master equation. A local Diff×Weyl anomaly could obstruct quantum nilpotency even if Corollary 8.6 is fully established.

Vertex versus dynamics. The two residual classes do not define two independent local gravitational dynamics. The parity-even class is the Weyl action direction; the parity-odd class is a theta direction, locally related by a canonical boundary transformation. The positive dynamical quotient is therefore one-dimensional.

The remaining classical bridge. The all-level smooth metric-to-$\mathcal{W}_{\text{geom}}$ identification is exact. Three sharper obligations remain: establish algebraic cylinder-mode exactness and the explicit $E/A/L$ curvature intertwiner; construct the full $SO(4,2)$-equivariant cyclic BV zero-mode transfer with its antifield-row inventory; and derive the chosen four-ghost polarization and normalized pairing from the residual BFV boundary problem. The direct Taub argument does not require first identifying a degree-three BGG representative, although that geometric comparison remains valuable. Extension to a chosen Hilbert or Krein completion is a separate analytic question. The exact ranks in Table 1 remain independent convention and regression checks, not the proof of algebraic-mode exactness.

10 Conclusion

The conventional free oscillator description of pure conformal gravity on $\mathbb{R} \times S^3$ is indefinite. Residual conformal reduction changes the question rather than re-signing those oscillators. The absolute residual complex carries a Cartan contraction which eliminates every noncentered compact degree. At the ghost number four selected by the centered polarization this turns the apparent infinite problem into the three sectors $N = 0, 1, 2$. The first two are acyclic; the last has exactly two classes and positive Gram matrix:

$$H^4_{\text{res,min}} = \text{span}\{[W^2_+], [W^2_-]\}, \qquad G_{\text{res}} = I_2.$$

Descent identifies them as the Weyl-square and Pontryagin vertex directions, and the quotient by variationally trivial bulk functionals leaves one positive dynamical direction.

The result is both stronger and narrower than a sampled positivity calculation. It is exhaustive for the stated minimal residual complex, but it concerns vertices rather than propagating particles. Its promotion to the full free pure-Weyl BV theory is controlled by the explicit hypotheses of Corollary 8.6. Quantum consistency and interaction descent are separate questions.

A Conventions

The cylinder radius is one and the Lorentzian signature is $(-,+,+,+)$. Compact energy is the eigenvalue of $D = i\partial_t$ in the oscillator convention. The compact algebra is $SO(4) \cong SU(2)_L \times SU(2)_R$. A label (j_L, j_R) has dimension $(2j_L+1)(2j_R+1)$. Positive chirality means the lowest Weyl primary $(2; 2, 0)$; parity exchanges j_L and j_R.

The normalized module convention is (39). The reduced curvature action (8) differs by an overall factor and sign from the common $-\int C^2$ oscillator convention. All moment-map comparisons derive the symplectic form and Taub kernel from the same action, so the overall action factor cancels. Lorentzian self-duality may insert a factor of i in the odd parity combination; no conclusion depends on that convention.

The residual generator degrees are $(0^7, -1^4, +1^4)$ and dual ghost degrees are their negatives. Ghost number and compact degree are independent. The selected absolute residual degree in this paper is ghost number four, fixed by (58); it is not ordinary ghost number zero. We call it a candidate physical degree only under the full residual gauging specified in Section 5.2.

B Exact rank certification

The one-particle matrices contain radicals. They are represented over

$$K = \mathbb{Q}(i, \sqrt{2}, \sqrt{3}, \sqrt{5}). \tag{120}$$

For a lower bound on rank, the calculation maps this field to $\mathbb{F}_{241}$ by

$$i \mapsto 64, \qquad \sqrt{2} \mapsto 22, \qquad \sqrt{3} \mapsto 56, \qquad \sqrt{5} \mapsto 103. \tag{121}$$

Each image obeys its defining square relation. A nonzero minor after this good-prime reduction is a nonzero characteristic-zero minor. Exact nilpotency provides the complementary upper bound

$$\operatorname{rank} d_{q-1} + \operatorname{rank} d_q \leq \dim C^q. \tag{122}$$

For the one-particle complex the modular lower bounds saturate this inequality. For the two-particle complex the two explicit independent kernel vectors provide the upper bound which the modular rank saturates. Thus the ranks quoted in Section 6 are exact characteristic-zero ranks, not probabilistic modular estimates.

C Claim-status and verification index

The residual computation used in the first draft was frozen at repository commit

`e928f257c25099eb534eb34109dfc1dc6a3127a1`.

The global BGG bridge was added in the subsequent theorem revision based at

`c471b99f5e3708e692b1c25238f6272c9e29b48f`.

The companion snapshot records both layers, the environment, hashes, expected-failure guards, and command lines. The complete paper verification is run with

```
python3 symbolic/verify_conformal_paper_free.py
```

Claim	Status	Evidence / boundary
Formal pure-Weyl detour identities	Exact	Branson–Gover theorem plus action-normalized factorization and finite scalar-block audits.
Global smooth deformation cohomology and curvature isomorphism	Exact smooth theorem	The flat BGG fine resolution and $H^\bullet(\mathbb{R}\times S^3)$ give $H^1_{\mathrm{def}}=H^2_{\mathrm{def}}=0$ and hence (23). This does not establish a completed Hilbert/Krein isomorphism, exactness on the algebraic positive-energy mode complex, or the BV/BFV zero-mode split.
Euclidean polynomial detour ranks, degrees 2–6	Exact finite certificate	Rational matrices verify kernels, ranks, and the 15 residual zero modes; they are an independent convention audit rather than the source of the global theorem.
On-shell Weyl module and stable generator coefficients	Exact abstract module construction	Character, tower inventory, invariant form, and cutoff-stable conformal brackets. Its geometric curvature intertwiner and algebraic metric potentials remain Conjecture 2.5.
Residual Cartan homotopy reduction	Theorem	Explicit homotopy ι_D/δ on every nonzero total degree; this is a quotient only in the full residual-gauging problem.
Centered vacuum and one-particle sectors	Exact	Semisimple Lie-algebra cohomology for the vacuum; cutoff-complete absolute rank certificate for one particle.
Centered two-particle sector and $G_{\mathrm{res}}=I_2$	Exact residual theorem	Independent relative and absolute calculations; empty incoming absolute space and exact rank 53.
Vertex descent and dynamical/topological split	Exact locally	Weight-four descent, Chern–Weil transgression, and Euler–Lagrange quotient. Global theta effects are retained.
Full local-plus-residual pure-Weyl BV result	Conditional	Requires the three parts of Field-Theoretic Hypothesis 8.5: algebraic cylinder realization, full equivariant cyclic BV transfer with row control, and the selected residual BFV polarization and normalization.
Interpretation as a physical state space	Choice-dependent	Requires a boundary and gauge principle that treats all fifteen generators, including D, as gauge rather than global charges.
Quantum anomaly freedom and interacting probability	Not claimed	Outside the scope of this paper.

References

[1] T. Branson and A. R. Gover, "The conformal deformation detour complex for the obstruction tensor," arXiv:math/0605192.

[2] A. R. Gover and L. J. Peterson, "The ambient obstruction tensor and the conformal deformation complex," Pacific J. Math. **226** (2006) 309–351, arXiv:math/0408229.

[3] A. R. Gover, P. Somberg, and V. Souček, "Yang–Mills detour complexes and conformal geometry," Commun. Math. Phys. **278** (2008) 307–327, arXiv:math/0606401.

[4] M. Benini, G. Musante, and A. Schenkel, "Green hyperbolic complexes on Lorentzian manifolds," Commun. Math. Phys. (2023), doi:10.1007/s00220-023-04807-5, arXiv:2207.04069.

[5] A. Čap, "Overdetermined systems, conformal geometry, and the BGG complex," in *Symmetries and Overdetermined Systems of Partial Differential Equations*, IMA Vol. Math. Appl. **144** (Springer, 2008) 1–24, arXiv:math/0610225.

[6] A. Čap, "Infinitesimal automorphisms and deformations of parabolic geometries," J. Eur. Math. Soc. **10** (2008) 415–437, arXiv:math/0508535.

[7] A. Čap and K. Hu, "Bounded Poincaré operators for twisted and BGG complexes," J. Math. Pures Appl. **179** (2023) 253–276, arXiv:2304.07185.

[8] K.-j. Hamada and S. Horata, "Conformal algebra and physical states in non-critical 3-brane on $\mathbb{R} \times S^3$," Prog. Theor. Phys. **110** (2003) 1169–1210, arXiv:hep-th/0307008.

[9] K.-j. Hamada, "Building blocks of physical states in a non-critical 3-brane on $\mathbb{R} \times S^3$," Int. J. Mod. Phys. A **20** (2005) 5353–5398, arXiv:hep-th/0402136.

[10] K.-j. Hamada, "Conformal field theory on $\mathbb{R} \times S^3$ from quantized gravity," Int. J. Mod. Phys. A **24** (2009) 3073–3110, arXiv:0811.1647.

[11] K.-j. Hamada, "BRST analysis of physical fields and states for 4D quantum gravity on $\mathbb{R} \times S^3$," Phys. Rev. D **86** (2012) 124006, arXiv:1202.4538.

[12] S. M. Kuzenko and M. Ponds, "Conformal geometry and (super)conformal higher-spin gauge theories," JHEP **05** (2019) 113, arXiv:1902.08010.

[13] M. Beccaria, X. Bekaert, and A. A. Tseytlin, "Partition function of free conformal higher spin theory," JHEP **08** (2014) 113, arXiv:1406.3542.

[14] N. Boulanger and M. Henneaux, "A derivation of Weyl gravity," Annalen Phys. **10** (2001) 935–964, arXiv:hep-th/0106065.

[15] G. Barnich, F. Brandt, and M. Henneaux, "Local BRST cohomology in gauge theories," Phys. Rept. **338** (2000) 439–569, arXiv:hep-th/0002245.

[16] G. Barnich, F. Brandt, and M. Henneaux, "Local BRST cohomology in the antifield formalism: I. General theorems," Commun. Math. Phys. **174** (1995) 57–92, arXiv:hep-th/9405109.

[17] G. Barnich, M. Henneaux, T. Hurth, and K. Skenderis, "Cohomological analysis of gauged-fixed gauge theories," Phys. Lett. B **492** (2000) 376–384, arXiv:hep-th/9910201.

[18] R. R. Metsaev, "BRST–BV approach to conformal fields," J. Phys. A **49** (2016) 175401, arXiv:1511.01836.

[19] I. Dneprov and M. Grigoriev, "Presymplectic BV–AKSZ formulation of conformal gravity," doi:10.1140/epjc/s10052-022-11082-6, arXiv:2208.02933.

[20] G. Barnich, X. Bekaert, and M. Grigoriev, "Notes on conformal invariance of gauge fields," J. Phys. A **48** (2015) 505402, arXiv:1506.00595.

[21] E. Altas and B. Tekin, "Linearization instability for generic gravity in AdS," Phys. Rev. D **97** (2018) 024028, arXiv:1705.10234.

[22] E. Altas and B. Tekin, "Second order perturbation theory in general relativity: Taub charges as integral constraints," Phys. Rev. D **99** (2019) 104078, arXiv:1903.11982.

[23] C. Chevalley and S. Eilenberg, "Cohomology theory of Lie groups and Lie algebras," Trans. Amer. Math. Soc. **63** (1948) 85–124.

[24] W. Fischler and S. Kundu, "Physical effects of the gravitational Θ-parameter," JHEP **05** (2017) 098, arXiv:1612.06010.